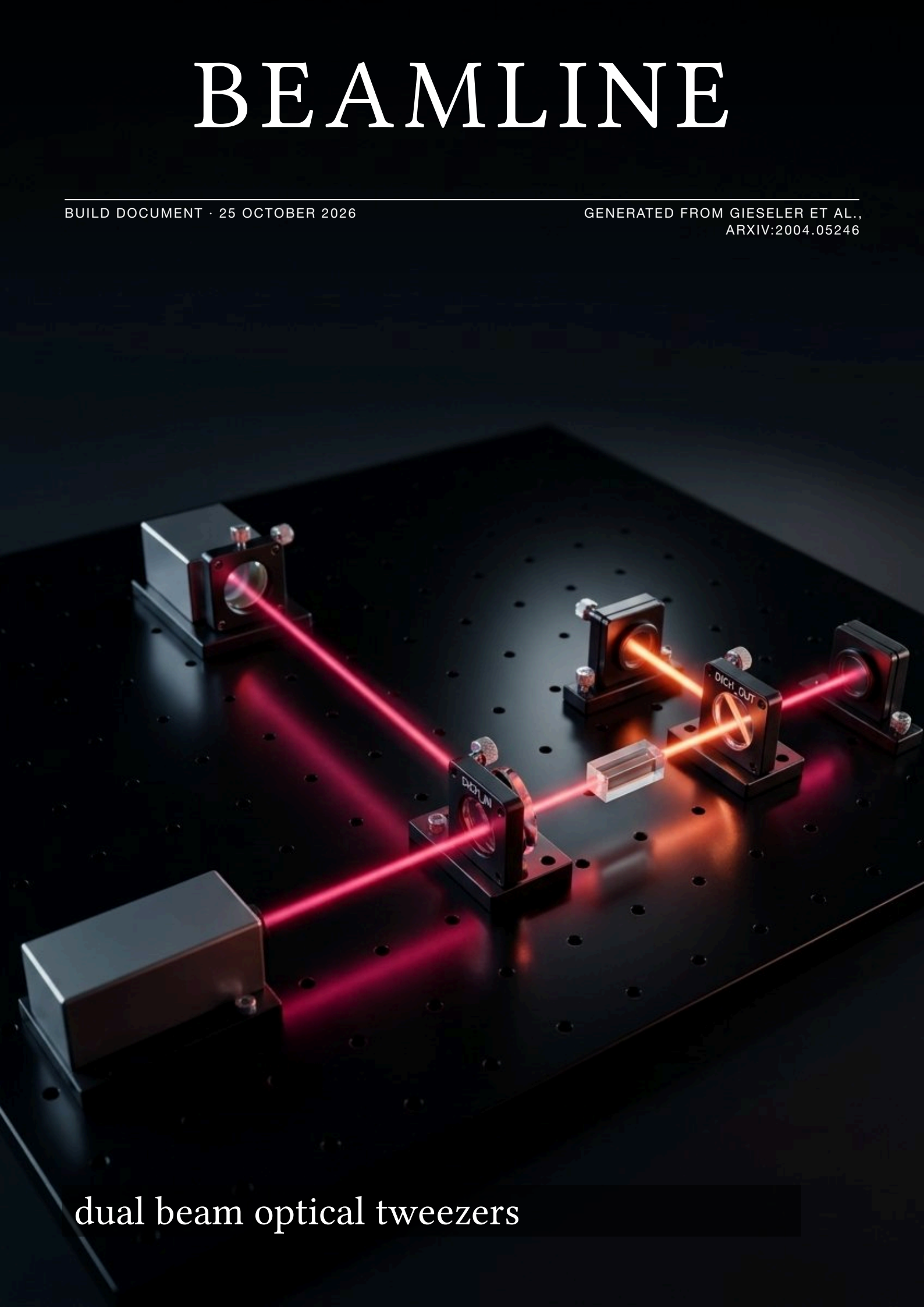
BEAMLINE

BUILD DOCUMENT · 25 OCTOBER 2026

GENERATED FROM GIESELER ET AL., ARXIV:2004.05246



dual beam optical tweezers

Overview

Dual-beam optical tweezers setup (after Gieseler et al., Fig. 5). A 532 nm trapping beam is split into two orthogonal polarizations by a half-wave plate (HWP) and polarizing beamsplitter (PBS1), steered independently by two gimbal mirrors (GM1, GM2), recombined at PBS2, and relayed by a 4f telescope (L1, L2) onto a dichroic mirror (DM) that sends the green trapping beams through objective O into the sample on the 3D stage. Top-down yellow LED illumination (LS, 590 nm) passes through condenser C, the sample, and objective O, transmits through DM, reflects off mirror M, and is focused by the tube lens (TL) onto the camera (Cam).

Dual-beam optical tweezers setup from Gieseler et al., Fig. 5. Both polarizations split at PBS1 (at [30, 139]) are steered by GM1 ([30, 214]) and GM2 ([105, 139]), recombined at PBS2 ([105, 214]), and relayed through the 4f system L1 ([180, 214], f=50 mm) and L2 ([280, 214], f=50 mm) onto the dichroic mirror DM ([350, 214], cutoff 560 nm). Both 532 nm trapping beams reflect upward through the objective O ([350, 281], f=40 mm), focus into the sample on the 3D stage ([350, 321]), are re-collimated by the condenser C ([350, 361], f=40 mm), and are safely deflected above C by DM_top ([350, 402]) into BD_trap ([295, 402]). Counter-propagating 590 nm illumination from LS ([350, 443]) transmits through DM_top, C, sample, O, and DM down to mirror M ([350, 144]), and through the tube lens TL ([287, 144], f=85 mm) onto the camera Cam ([203, 144]).

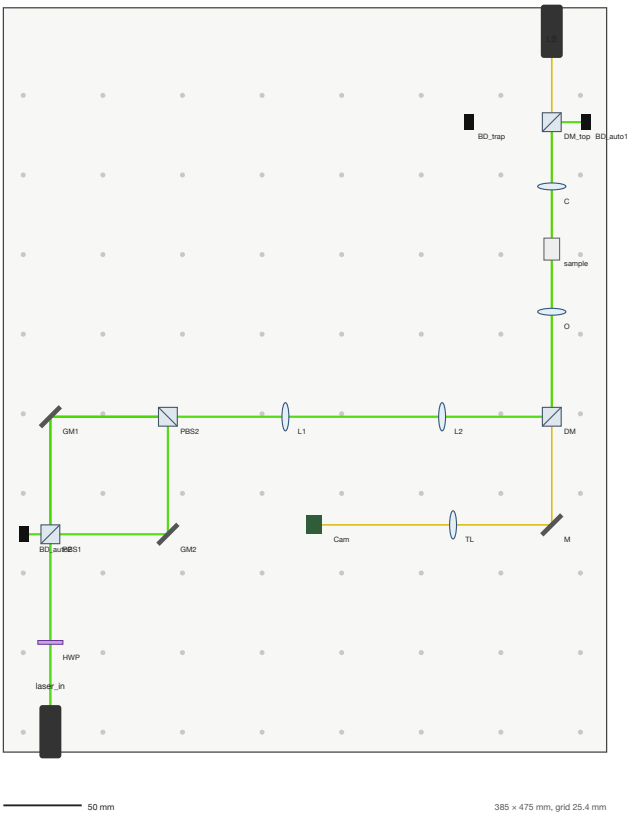
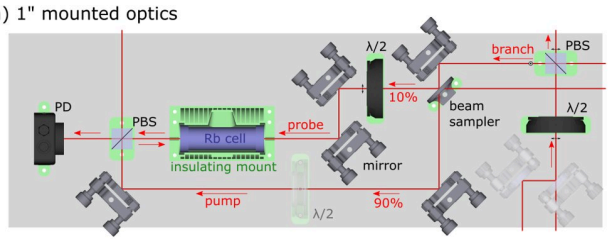


Figure 1: Plan view. Beams coloured by wavelength (infrared shown dark magenta), width ∝ log power; dashed = leaves the plate; grey dots = table holes.

Key specifications

Quantity	Value
Wavelengths	532 nm, 590 nm
Input power	110 mW from 2 source(s)
Delivered (fibers + detectors)	8.5 mW
Elements / beam segments	18 / 25
Baseplate	385 × 475 × 25.4 mm, 6 printed tile(s)

Source figure



Gieseler et al., arXiv:2004.05246 (CC BY 4.0)

Verification

PASS score 80 / 100 (Beamline tracer: 2D ray + Gaussian beam + Jones polarization)

Targets

Target	Status
met: Cam: 8.500 mW (need ≥ 4.000)	met

Checks

Severity	Check	Message
WARNING	collision	PBS1 and BD_auto2 footprints overlap (15.0 mm apart)
WARNING	collision	O and sample footprints overlap (40.0 mm apart)
WARNING	collision	sample and C footprints overlap (40.0 mm apart)
INFO	target	met: Cam: 8.500 mW (need ≥ 4.000)

Bill of materials

Purchased parts

Part	Description	Vendor	Qty	Unit	Line
BB1-E02	Ø1" broadband dielectric mirror, 400-750 nm	Thorlabs	3	$80.00	$240.00
DMLP567	dichroic mirror (DM (longpass dichroic, cutoff 560 nm, reflects < 560 nm: reflects 532 nm trapping beam from +x into +y toward objective O, transmits 590 nm illumination from -y into -y toward mirror M!))	Thorlabs	2	$190.00	$380.00
KM100	kinematic mirror mount, Ø1"	Thorlabs	5	$48.00	$240.00
KM100CP	kinematic mount for 10 mm cubes	Thorlabs	2	$70.00	$140.00
LA1131-A	lens (L1 (4f relay lens 1, f = 50 mm, separated by f1+f2 = 100 mm from L2 -> collimated 1:1 relay of GM1/GM2 onto objective back aperture!))	Thorlabs	2	$35.00	$70.00
LA1509-A	lens (Tube lens (f = 85 mm at [287, 144], focuses imaging beam onto camera Cam at [203, 144]))	Thorlabs	1	$35.00	$35.00
LB1	beam block / dump, black anodized	Thorlabs	3	$95.00	$285.00
LMR1	lens mount, Ø1" optics	Thorlabs	5	$18.00	$90.00
PBS251	polarizing beamsplitter (PBS1 (splits into transmitted H -> GM1 and reflected V -> GM2 along +x, out_direction_deg = 0.0 -> angle_deg = 180.0, diagonal normal at 315/135 deg, matching figure diagonal /))	Thorlabs	2	$290.00	$580.00
RSP05	rotation mount, Ø1/2" optics	Thorlabs	1	$95.00	$95.00
WPH05M-532	half-wave plate (lambda/2 (50:50 split at PBS1, H -> D at 22.5 deg; reflection at GM1/GM2 flips Ex cleanly for recombination at PBS2: GM1 sends H through PBS2, GM2 sends V reflected by PBS2 along +x -> 100% recombined!))	Thorlabs	1	$310.00	$310.00
Coherent Verdi V5	532 nm trapping laser input (532 nm, 100 mW)		1	—	—
Olympus Condenser	lens (Condenser C (f = 40 mm at y = 361 mm, 40 mm above sample at y = 321 mm -> re-collimates trapping beam and focuses illumination LS into sample, which O then re-collimates!))		1	—	—
Olympus Plan N 40x	lens (Objective O (collimated relay / microscope objective pair with condenser C: f = 40 mm separated by 80 mm at y = 281 and y = 361 mm with sample at midpoint y = 321 mm!))		1	—	—
Sample Chamber	vapor cell (3D sample stage (microfluidic chamber / trapping sample cell at focus y = 321 mm, length 20 mm so footprint 20x25 clears O at 281 and C at 361; transmission 0.85 for scattering/absorption at bead/water interface!))		1	—	—
Thorlabs CS165MU	photodiode (Camera (CMOS imaging sensor at [203, 144], 84 mm after tube lens TL!))		1	—	—
Thorlabs M590L4	Illumination light source (590 nm LED/lamp, LS -> C -> sample -> O -> DM -> M -> TL -> Cam) + trapping dump (590 nm, 10 mW)		1	—	—

Estimated total: $2,508.40

list-price estimates, USD, excluding lasers; verify with vendors; no price for Olympus Condenser, Olympus Plan N 40x, Sample Chamber, Thorlabs CS165MU

Hardware

Part	Description	Qty	Unit
SH25S050	1/4-20 x 1/2" cap screw (25 pack -> per piece)	70	$0.40
SH8S100	8-32 x 1" cap screw (per piece)	18	$0.30
DOWEL-2x8	Ø2 x 8 mm steel dowel pin (per piece)	36	$0.15
M3x25+NUT	M3 x 25 mm cap screw + nut (tile lap joints)	23	$0.20

3D-printed parts

Part	STL file	cm³	Print h	PLA g
baseplate_r1c1	stl/baseplate_r1c1.stl	825.9	22.4	334
baseplate_r2c1	stl/baseplate_r2c1.stl	697.3	19.2	285
baseplate_r3c1	stl/baseplate_r3c1.stl	601.1	16.2	241
baseplate_r1c2	stl/baseplate_r1c2.stl	915.6	24.5	365
baseplate_r2c2	stl/baseplate_r2c2.stl	775.1	21.2	316
baseplate_r3c2	stl/baseplate_r3c2.stl	671.2	18.1	270

Assembly

Step 1 **Print the baseplate** — Print 6 file(s) from stl/ flat, top side up: PLA, 20 % infill, 3 walls (≈ 121.6 h, ≈ 1811 g).

Step 2 **Join the tiles** — Lay the tiles out by their r/c names, overlap the 20 mm half-lap joints and fasten with 23 M3 × 25 bolts + nuts.

Step 3 **Press in the dowel pins** — Press 36 Ø2 × 8 mm pins into the paired pin holes; they key each mount's orientation.

Step 4 **Bolt to the table** — Fix the plate with 70 1/4-20 (or M6) cap screws through the counterbored holes (2-grid pitch, half a grid in from the edges).

Step 5 **HWP** — Mount lambda/2 (RSP05, WPH05M-532) at site 1, (30.0, 70.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Set the fast axis to 22.5° from horizontal (H).

Step 6 **PBS1** — Mount PBS1 (KM100CP, PBS251) at site 2, (30.0, 139.0) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Outputs: laser_in.t → 90.0° (50 mW); laser_in.r → 360.0° (50 mW); laser_in.r.r.r → 180.0° (50 mW)

Step 7 **GM1** — Mount GM1 (BB1-E02, KM100) at site 3, (30.0, 214.0) mm; orientation 315.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: laser_in.t → 360.0° (50 mW); laser_in.r.r → 270.0° (50 mW)

Step 8 **PBS2** — Mount PBS2 (KM100CP, PBS251) at site 4, (105.0, 214.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. Outputs: laser_in.t.t → 360.0° (50 mW); laser_in.r.r → 180.0° (50 mW)

Step 9 **L1** — Mount L1 (LA1131-A, LMR1) at site 5, (180.0, 214.0) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. f = 50 mm, 75.0 mm after the previous element on laser_in.t.t.

Step 10 **L2** — Mount L2 (LA1131-A, LMR1) at site 6, (280.0, 214.0) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. f = 50 mm, 100.0 mm after the previous element on laser_in.t.t.

Step 11 **DM** — Mount DM (DMLP567, KM100) at site 7, (350.0, 214.0) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Outputs: laser_in.t.t.r → 90.0° (50 mW); LS.t.t → 270.0° (8.5 mW). Reflects below 560 nm.

Step 12 **O** — Mount Objective O (LMR1, Olympus Plan N 40x) at site 8, (350.0, 281.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. f = 40 mm, 67.0 mm after the previous element on laser_in.t.t.r.

Step 13 **sample** — Mount 3D sample stage (Sample Chamber) at site 9, (350.0, 321.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below.

Step 14 **C** — Mount Condenser C (LMR1, Olympus Condenser) at site 10, (350.0, 361.0) mm; orientation 90.0°; keyed by pins, one 8-32 screw from below. f = 40 mm, 40.0 mm after the previous element on laser_in.t.t.r.

Step 15 **DM_top** — Mount Upper shortpass/longpass separator (DMLP567, KM100) at site 11, (350.0, 402.0) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Outputs: laser_in.t.t.r.r → 360.0° (42.5 mW); LS.t → 270.0° (10 mW). Reflects below 560 nm.

Step 16 **BD_auto1** — Mount auto dump (LB1) at site 12, (370.0, 402.0) mm; orientation 360.0°; keyed by pins, one 8-32 screw from below. Blocks: laser_in.t.t.r.r (42.5 mW, 532 nm)

Step 17 **GM2** — Mount GM2 (BB1-E02, KM100) at site 13, (105.0, 139.0) mm; orientation 135.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: laser_in.r → 90.0° (50 mW)

Step 18 **BD_auto2** — Mount auto dump (LB1) at site 14, (15.0, 139.0) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Blocks: laser_in.r.r.r (50 mW, 532 nm)

Step 19 **M** — Mount Mirror M (BB1-E02, KM100) at site 15, (350.0, 145.0) mm; orientation 135.0°; keyed by pins, one 8-32 screw from below. Steer the beam to: LS.t.t → 180.0° (8.5 mW)

Step 20 **TL** — Mount Tube lens (LA1509-A, LMR1) at site 16, (287.0, 145.0) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. f = 85 mm, 63.0 mm after the previous element on LS.t.t.

Step 21 **Cam** — Mount Camera (Thorlabs CS165MU) at site 17, (203.0, 145.0) mm; orientation 180.0°; keyed by pins, one 8-32 screw from below. Expect ≈ 8.5 mW on the detector.

Step 22 **BD_trap** — Mount Upper dump / condenser stop for transmitted 532… (LB1) at site 18, (295.0, 402.0) mm; orientation 0.0°; keyed by pins, one 8-32 screw from below. Blocks: (nothing in this state)

Alignment procedure

Align in beam order with an iris and an IR card; adjust one knob at a time and keep the best reading.

Order: HWP → GM1 → L1 → L2 → O → C → GM2 → M → TL

No tuner run attached: use the design values listed in the assembly steps.

Expected readings

- expect ≈ 8.5 mW at Cam

Laser safety

Wavelengths: 532 nm, 590 nm. **Maximum source power on the table:** 100 mW.

Wavelength	Band	Eyewear (generic guidance)
532 nm	visible	OD ≥ 3 at 532 nm (generic; confirm with your laser safety officer)
590 nm	visible	OD ≥ 3 at 590 nm (generic; confirm with your laser safety officer)

Beam dumps

Dump	Blocks
BD_auto1	laser_in.t.t.r.r (42.5 mW, 532 nm)
BD_auto2	laser_in.r.r.r (50 mW, 532 nm)
BD_trap	(nothing in the default state)

Verify all beam dumps before full power. Align at the lowest usable power, never look along the beam, remove watches and rings, keep the beam below eye level, and enclose invisible (IR) beams.

Credits

- Layout concept and “paint by numbers” baseplates: PyOpticL (code-to-CAD optics).
- Source paper: Gieseler et al., arXiv:2004.05246 — license: CC BY 4.0
- Designed by Gemini `gemini-4-argon` agents, verified by the Beamline tracer.
- Prices are list-price estimates; part numbers should be checked against current vendor catalogs.